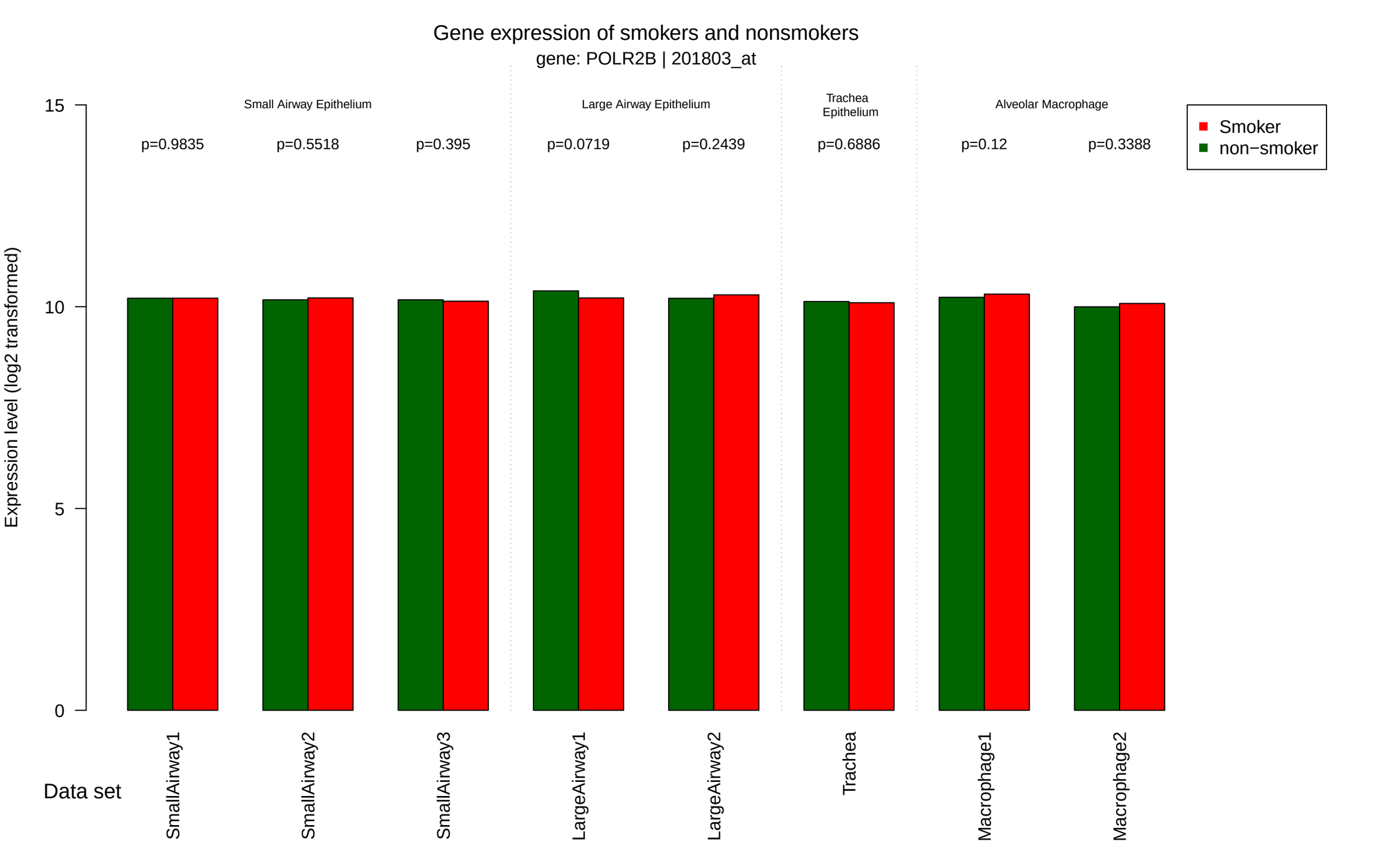
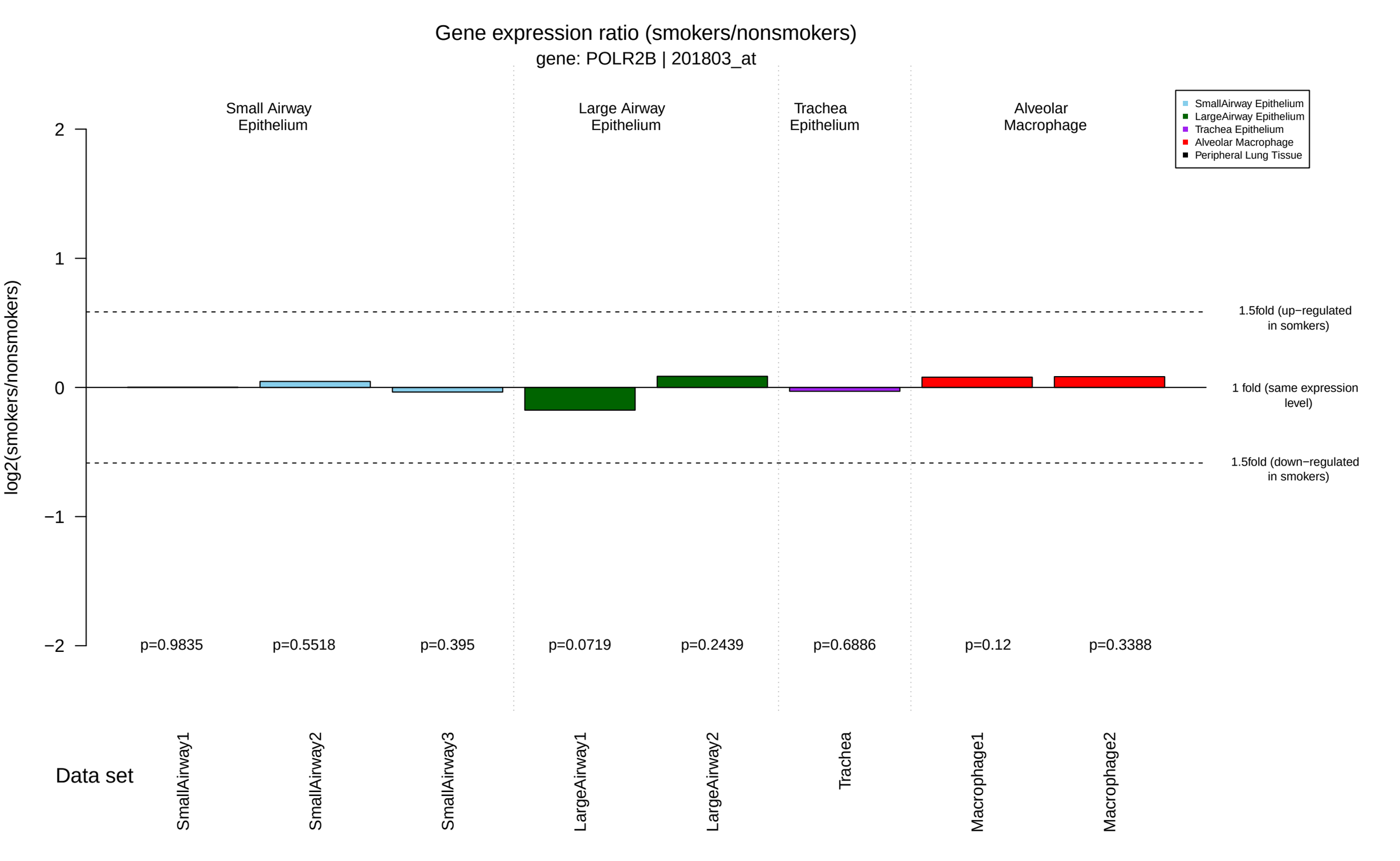
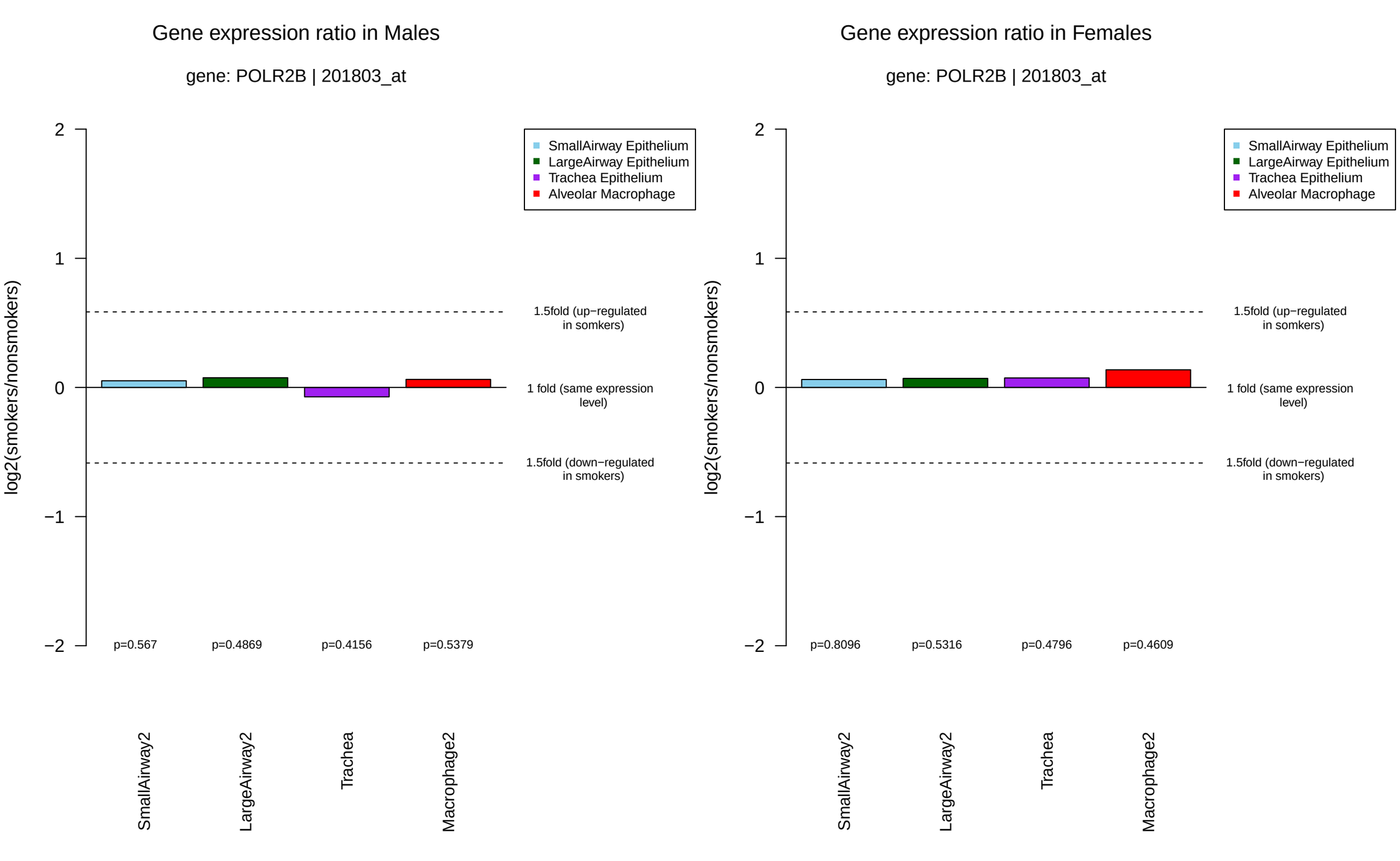
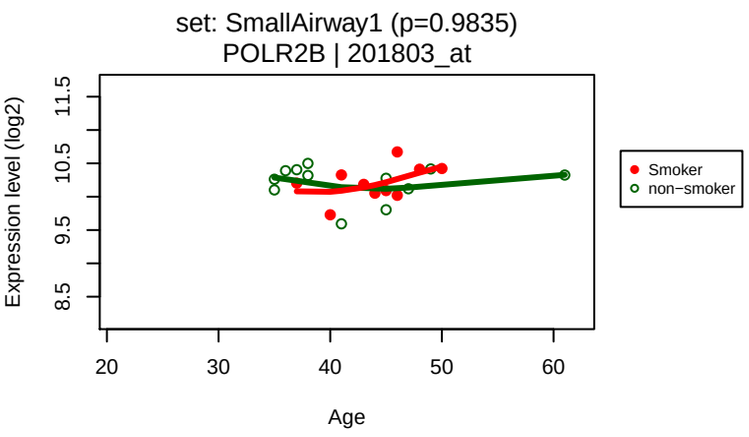
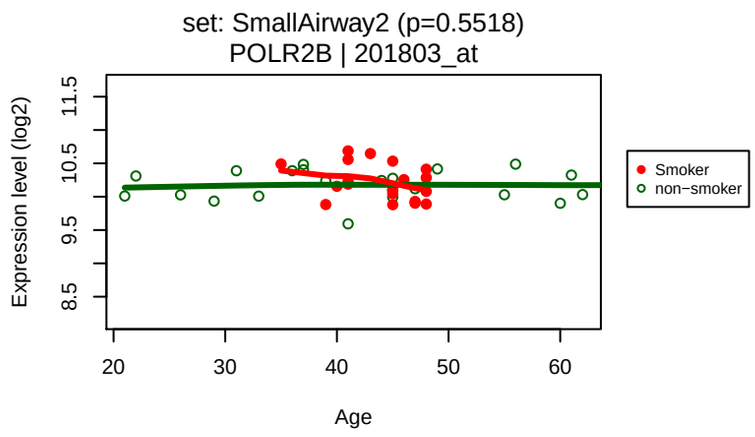
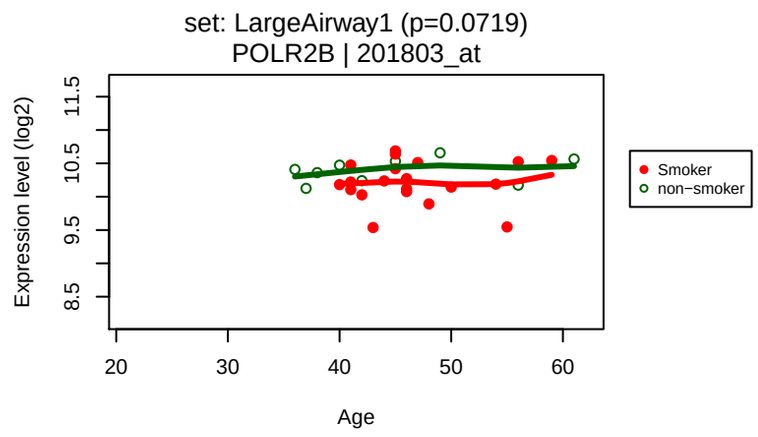

Gene expression of smokers and nonsmokers
gene: POLR2B | 201803_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9835
p=0.5518
p=0.395
p=0.0719
p=0.2439
p=0.6886
p=0.12
p=0.3388
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

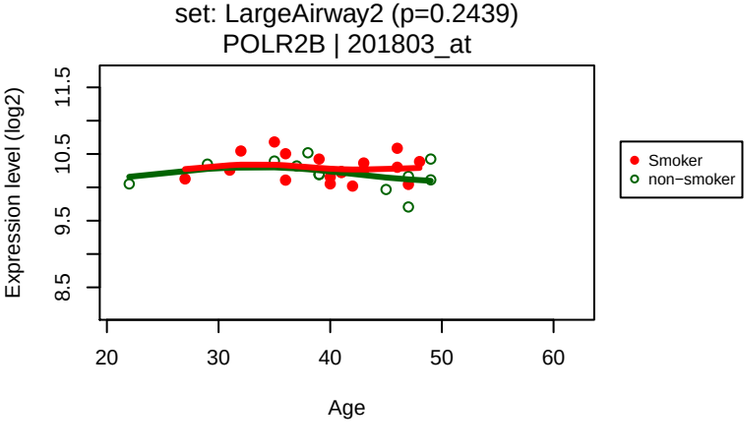

Gene expression ratio (smokers/nonsmokers)
gene: POLR2B | 201803_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9835
p=0.5518
p=0.395
p=0.0719
p=0.2439
p=0.6886
p=0.12
p=0.3388
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

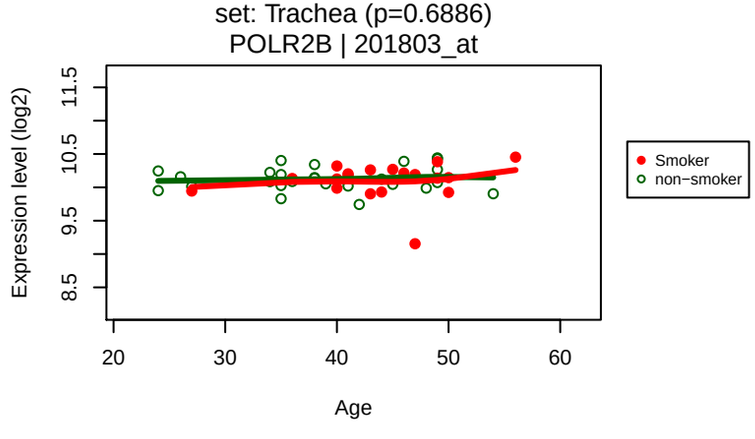

Gene expression ratio in Males
gene: POLR2B | 201803_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.567
p=0.4869
p=0.4156
p=0.5379
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR2B | 201803_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8096
p=0.5316
p=0.4796
p=0.4609
SmallAirway2
LargeAirway2
Trachea
Macrophage2

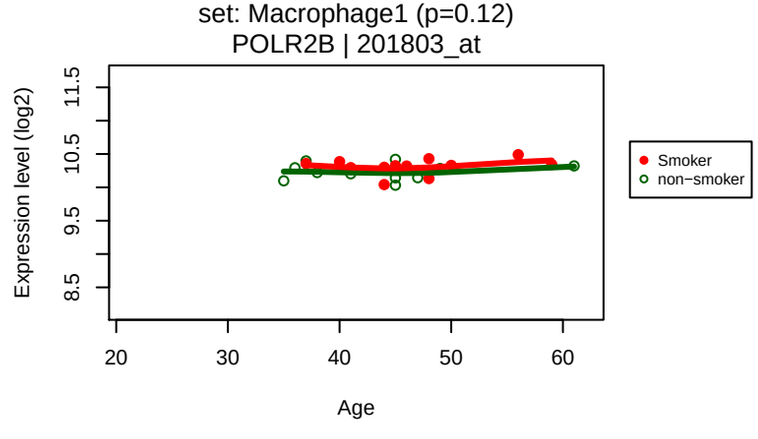

set: SmallAirway1 (p=0.9835)
POLR2B | 201803_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker

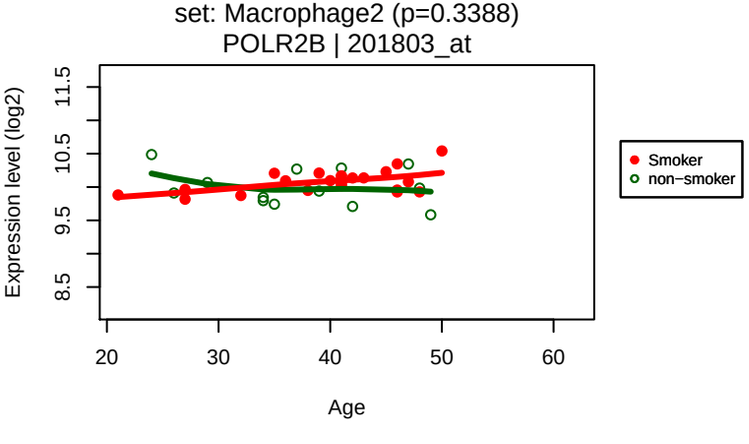

set: SmallAirway2 (p=0.5518)
POLR2B | 201803_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker

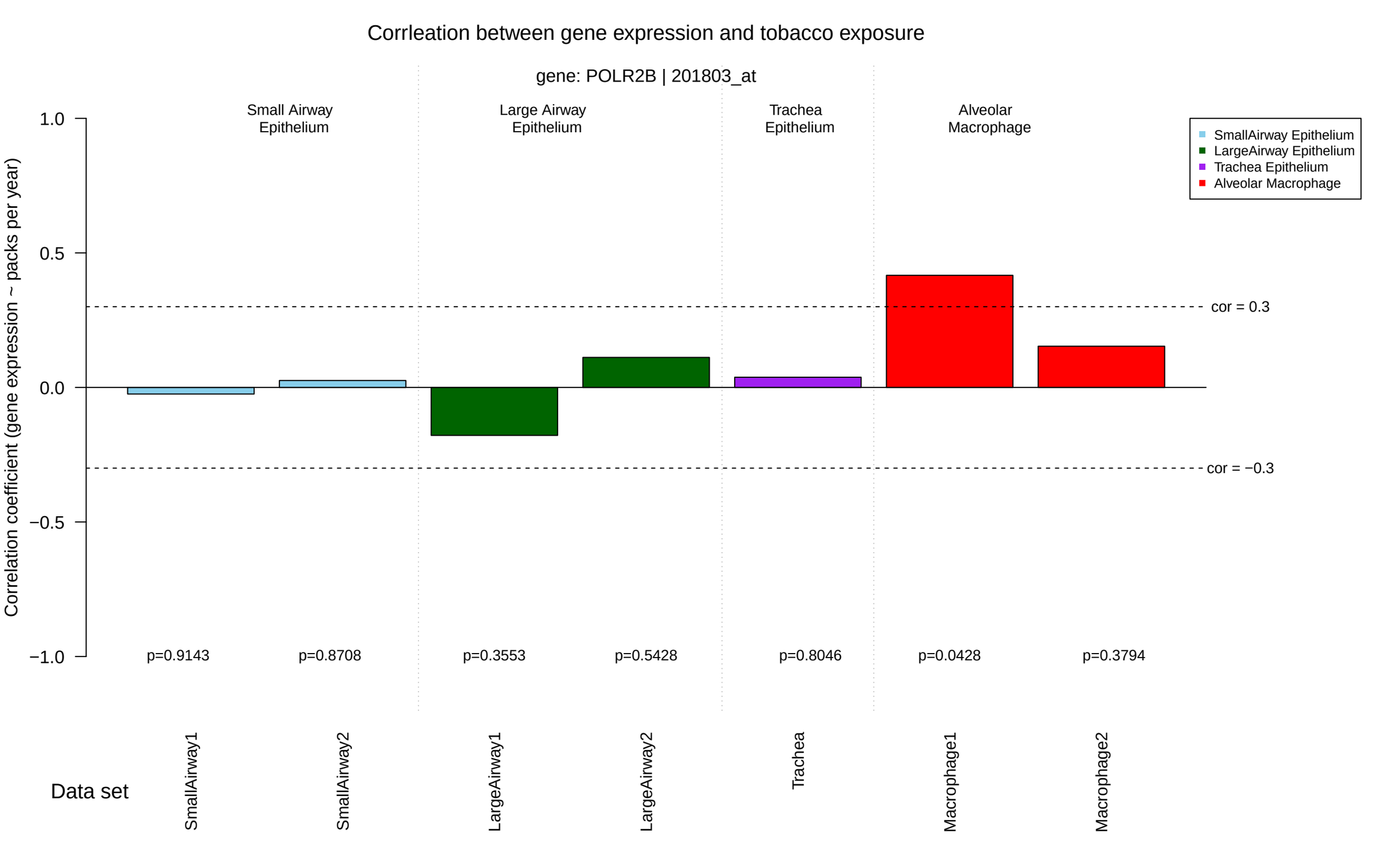

set: LargeAirway1 (p=0.0719)
POLR2B | 201803_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2439)
POLR2B | 201803_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.6886)
POLR2B | 201803_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.12)
POLR2B | 201803_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3388)
POLR2B | 201803_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR2B | 201803_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9143
p=0.8708
p=0.3553
p=0.5428
p=0.8046
p=0.0428
p=0.3794
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2